CENTRAL BOARD OF SECONDARY EDUCATION

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COMPUTER SCIENCE (083)

PROJECT FILE

On

LIBRARY MANGEMENT SYSTEM USING PICKLE MODULE

SUBMITTED BY :

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4.		

CLASS : XII
SECTION : SCIENCE

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I also extend my thanks to a number of teachers, my classmates and friends who helped me to complete this project file successfully.

NAME :

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CERTIFICATE

This is to certify that,

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Student of Class XII, J.N.V Tarikhet Almora has completed the PROJECT FILE during the academic year 2026-27 towards partial fulfilment of credit for the Computer Science Project evaluation of CBSE and submitted satisfactory report, as compiled in the following pages, under my supervision.

Internal Examiner
Signature

External Examiner
Signature

PREFACE

*This project report details the development of a **Library Management System** using Python. The project is designed to automate the management of a school library, allowing the librarian to perform operations such as adding books, registering students, issuing and returning books, and tracking each student's issued and returned books. The project highlights Python's powerful data-handling capabilities, especially using the Pickle module for data persistence. The project aims to simplify and improve the efficiency of library operations.*

I also feel that it is the right time to thank my teacher, who guided me during my project. I also feel that it is the right time to thank my parents, friends and my project partners for their constant assistance, cooperation extended by them time to time.

Table of Contents

About Project 5
Introduction: 5
Programming Language: 5
Modules Used: 5
How to Run the Program: 5
System Features 5
1. Librarian Authentication 5
2. Student Registration 5
3. Book Management 5
4. Book Issuing and Returning 5
5. Tracking Issued and Returned Book 6
User Interaction: 6
1. Librarian Login: 6
2. Menu Options: 6
3. Student Registration: 6
4. Book Issuing: 6
5. Book Return: 6
Flow Chart 7
Project Code 8
Output 16
Technological requirements 19
Software 19
Hardware 19
References 20

About Project

Introduction:

The Library Management System is a Python-based project designed to automate library operations. This system allows the librarian to add and remove books, register students, issue books, and manage book returns. The system also tracks how many books a student has issued or returned. Data is saved using Python's Pickle module, ensuring the information persists between program runs.

Programming Language:

The project is implemented using the Python programming language, known for its simplicity and versatility. Python's ease of use makes it an ideal choice for developing interactive and enjoyable games.

Modules Used:

1. Pickle Module: To store and load library and student data.
2. UUID Module: To generate unique IDs for students.
3. OS Module: To handle file operations.

How to Run the Program:

1. Open the terminal or command prompt.
2. Navigate to the directory where the Python script is saved.
3. Run the script using the command:
   python library_management_system.py

The system will prompt the user for librarian authentication before allowing any operations.

System Features

1. Librarian Authentication

Only the librarian with a valid username and password can access the system.

2. Student Registration

Students are registered with their name, roll number, class, section, gender, and date of birth. Each student is assigned a unique student ID.

3. Book Management

Books can be added and removed by the librarian.

4. Book Issuing and Returning

Registered students can issue books. The system ensures that:

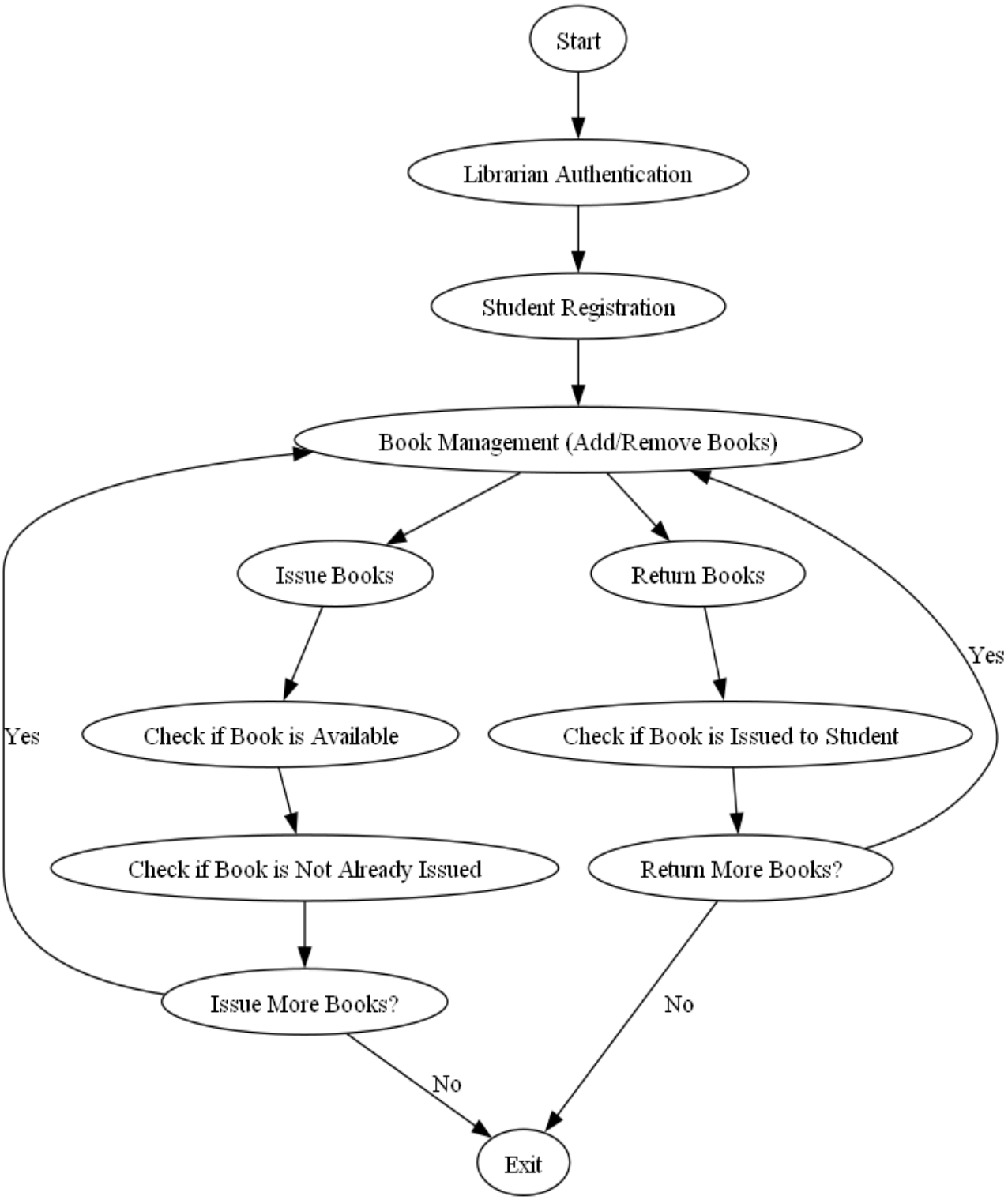
- A student can issue a maximum of 10 books at a time.
- The same book cannot be issued twice unless it is returned.
- Only books available in the library can be issued. After issuing a book, the system prompts if the student wants to issue another book.

5. **Tracking Issued and Returned Book**
   The system tracks the number of books issued and returned by each student.

User Interaction:

1. **Librarian Login:**
   The librarian needs to authenticate with a username and password to access the system.
2. **Menu Options:**
   The system displays a menu with options such as registering students, adding books, issuing books, etc.
3. **Student Registration:**
   The librarian enters the student's details, and the system generates a unique student ID.
4. **Book Issuing:**
   The librarian enters the student ID and book ID. The system checks if the book is available and issues it to the student.
5. **Book Return:**
   The librarian enters the student ID and book ID to return the book.

Flow Chart


Start
Librarian Authentication
Student Registration
Book Management (Add/Remove Books)
Issue Books
Return Books
Yes
Yes
Check if Book is Available
Check if Book is Issued to Student
Check if Book is Not Already Issued
Return More Books?
Issue More Books?
No
No
Exit

Project Code

```
import pickle
import os
import uuid  # For generating unique student IDs

# Files to store library and student data
library_data_file = 'library_data.pkl'
students_data_file = 'students_data.pkl'

# Librarian credentials (can be hardcoded for simplicity)
LIBRARIAN_USERNAME = "Pankaj"
LIBRARIAN_PASSWORD = "qwer"

# Function to load data from a pickle file
def load_data(file):
    if os.path.exists(file):
        with open(file, 'rb') as f:
            return pickle.load(f)
    else:
        return {}



# Function to save data to a pickle file
def save_data(data, file):
    with open(file, 'wb') as f:
        pickle.dump(data, f)



# Function to register students with extended details and auto-generated ID
def register_student():
    students = load_data(students_data_file)

    name = input("Enter Student Name: ")
```

```
    roll = input("Enter Roll Number: ")
    student_class = input("Enter Class: ")
    section = input("Enter Section: ")
    gender = input("Enter Gender (M/F): ")
    dob = input("Enter Date of Birth (DD-MM-YYYY): ")

    # Generate a unique student ID
    student_id = str(uuid.uuid4())[:8]  # Using first 8 characters of UUID as student ID

    # Register the student with details
    students[student_id] = {
        'name': name,
        'roll': roll,
        'class': student_class,
        'section': section,
        'gender': gender,
        'dob': dob,
        'issued_books': [],
        'books_returned': 0
    }

    save_data(students, students_data_file)
    print(f"Student '{name}' registered successfully with Student ID: {student_id}")


# Function to authenticate librarian
def authenticate_librarian():
    username = input("Enter librarian username: ")
    password = input("Enter librarian password: ")

    if username == LIBRARIAN_USERNAME and password == LIBRARIAN_PASSWORD:
        print("Librarian authenticated successfully!")
        return True
    else:
```

```
        print("Authentication failed! Invalid username or password.")
        return False



# Function to add a book to the library
def add_book(book_id, title, author):
    library = load_data(library_data_file)
    if book_id in library:
        print(f"Book with ID {book_id} already exists!")
    else:
        library[book_id] = {'title': title, 'author': author, 'is_issued': False, 'issued_to': None}
        save_data(library, library_data_file)
        print(f"Book '{title}' added successfully!")



# Function to remove a book from the library
def remove_book(book_id):
    library = load_data(library_data_file)
    if book_id in library:
        del library[book_id]
        save_data(library, library_data_file)
        print(f"Book ID {book_id} removed successfully!")
    else:
        print(f"Book with ID {book_id} not found!")



# Function to issue a book to a registered student, checking conditions
def issue_book(student_id):
    library = load_data(library_data_file)
    students = load_data(students_data_file)

    if student_id not in students:
        print(f"Student with ID {student_id} is not registered!")
        return
```

```
    student = students[student_id]

    # Check if the student has already issued the maximum of 10 books
    if len(student['issued_books']) >= 10:
        print(f"Student {student['name']} has already issued the maximum of 10 books!")
        return

    while True:
        book_id = input("Enter Book ID to issue: ")

        if book_id not in library:
            print(f"Book with ID {book_id} not found!")
            return

        if library[book_id]['is_issued']:
            print(f"Book ID {book_id} is already issued to {library[book_id]['issued_to']}!")
            return

        if book_id in student['issued_books']:
            print(f"Student {student['name']} has already issued this book.")
            return

        # Issue the book
        library[book_id]['is_issued'] = True
        library[book_id]['issued_to'] = student['name']
        student['issued_books'].append(book_id)

        save_data(library, library_data_file)
        save_data(students, students_data_file)

        print(f"Book ID {book_id} issued successfully to {student['name']}!")

        # Ask if the student wants to issue more books
```

```
        issue_more = input("Do you want to issue another book? (Y/N): ").strip().upper()
        if issue_more != 'Y':
            break



# Function to return a book by a student
def return_book(book_id, student_id):
    library = load_data(library_data_file)
    students = load_data(students_data_file)

    if book_id not in library:
        print(f"Book with ID {book_id} not found!")
        return

    if not library[book_id]['is_issued']:
        print(f"Book ID {book_id} is not issued!")
        return

    if student_id not in students:
        print(f"Student with ID {student_id} not registered!")
        return

    student = students[student_id]

    if book_id not in student['issued_books']:
        print(f"Student {student['name']} did not issue this book.")
        return

    # Return the book
    library[book_id]['is_issued'] = False
    library[book_id]['issued_to'] = None
    student['issued_books'].remove(book_id)
    student['books_returned'] += 1
```

```
    save_data(library, library_data_file)
    save_data(students, students_data_file)

    print(f"Book ID {book_id} returned successfully by {student['name']}!")



# Function to display all books
def display_books():
    library = load_data(library_data_file)
    if library:
        for book_id, book_info in library.items():
            status = 'Issued' if book_info['is_issued'] else 'Available'
            issued_to = book_info['issued_to'] if book_info['is_issued'] else 'N/A'
            print(f"Book ID: {book_id}, Title: {book_info['title']}, Author: {book_info['author']}, Status: {status}, Issued To: {issued_to}")
    else:
        print("No books in the library.")



# Function to display all registered students
def display_students():
    students = load_data(students_data_file)
    if students:
        print("Registered Students:")
        for student_id, student_info in students.items():
            print(f"Student ID: {student_id}, Name: {student_info['name']}, Class: {student_info['class']}, Section: {student_info['section']}, Issued Books: {len(student_info['issued_books'])}, Books Returned: {student_info['books_returned']}")
    else:
        print("No students registered.")



# Main Program
if __name__ == "__main__":
    if authenticate_librarian():  # Only proceed if librarian is authenticated
```

```
while True:
    print("\nLibrary Management System")
    print("1. Register Student")
    print("2. Add Book")
    print("3. Remove Book")
    print("4. Issue Book")
    print("5. Return Book")
    print("6. Display Books")
    print("7. Display Students")
    print("8. Exit")

    choice = input("Enter your choice: ")

    if choice == '1':
        register_student()

    elif choice == '2':
        book_id = input("Enter Book ID: ")
        title = input("Enter Book Title: ")
        author = input("Enter Book Author: ")
        add_book(book_id, title, author)

    elif choice == '3':
        book_id = input("Enter Book ID to remove: ")
        remove_book(book_id)

    elif choice == '4':
        student_id = input("Enter Student ID: ")
        issue_book(student_id)

    elif choice == '5':
        book_id = input("Enter Book ID to return: ")
        student_id = input("Enter Student ID: ")
        return_book(book_id, student_id)
```

```
elif choice == '6':
    display_books()

elif choice == '7':
    display_students()

elif choice == '8':
    print("Exiting...")
    break

else:
    print("Invalid choice! Please try again.")
```

Output

```
Enter librarian username: Pankaj
Enter librarian password: qwer
Librarian authenticated successfully!

Library Management System
1. Register Student
2. Add Book
3. Remove Book
4. Issue Book
5. Return Book
6. Display Books
7. Display Students
8. Exit
```

```
Library Management System
1. Register Student
2. Add Book
3. Remove Book
4. Issue Book
5. Return Book
6. Display Books
7. Display Students
8. Exit
Enter your choice: 2
Enter Book ID: b01
Enter Book Title: English
Enter Book Author: OK cooper
Book 'English' added successfully!
```

```
Library Management System
1. Register Student
2. Add Book
3. Remove Book
4. Issue Book
5. Return Book
6. Display Books
7. Display Students
8. Exit
Enter your choice: 6
Book ID: b01, Title: English, Author: OK cooper, Status: Available,
Issued To: N/A
```

```
Enter your choice: 1
Enter Student Name: Rohan
Enter Roll Number: 12
Enter Class: XII
Enter Section: SC
Enter Gender (M/F): M
Enter Date of Birth (DD-MM-YYYY): 10-04-2001
Student 'Rohan' registered successfully with Student ID: 01b8b3b9
```

```
Library Management System
1. Register Student
2. Add Book
3. Remove Book
4. Issue Book
5. Return Book
6. Display Books
7. Display Students
8. Exit
Enter your choice: 7
Registered Students:
Student ID: 01b8b3b9, Name: Rohan, Class: XII, Section: SC, Issued
Books: 0, Books Returned: 0
```

```
Library Management System
1. Register Student
2. Add Book
3. Remove Book
4. Issue Book
5. Return Book
6. Display Books
7. Display Students
8. Exit
Enter your choice: 4
Enter Student ID: 01b8b3b9
Enter Book ID to issue: b01
Book ID b01 issued successfully to Rohan!
Do you want to issue another book? (Y/N): y
Enter Book ID to issue: b01
Book ID b01 is already issued to Rohan!
```

```
Library Management System
1. Register Student
2. Add Book
3. Remove Book
4. Issue Book
5. Return Book
6. Display Books
7. Display Students
8. Exit
Enter your choice: 5
Enter Book ID to return: b01
Enter Student ID: 01b8b3b9
Book ID b01 returned successfully by Rohan!
```

```
Library Management System
1. Register Student
2. Add Book
3. Remove Book
4. Issue Book
5. Return Book
6. Display Books
7. Display Students
8. Exit
Enter your choice: 7
Registered Students:
Student ID: 01b8b3b9, Name: Rohan, Class: XII, Section: SC, Issued
Books: 0, Books Returned: 1
```

Technological requirements

Software

1. Python
2. VS Code

Hardware

1. Laptop/PC

References

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